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FIRST SEMESTER UG DEGREE EXAMINATION, NOVEMBER 2025

(CUFYUGP)

MAT1MN104-Mathematical Logic, Set Theory and Combinatorics

2024 Admissions

Maximum Time :2 Hours	**Maximum Marks :70**

Section A

All Question can be answered. Each Question carries 3 marks (Ceiling : 24 Marks)

1	Write the converse, inverse, and contrapositive of "If London is in France, then Paris is in England."
2	Rewrite the sentence "The product of any two real numbers x and y is positive. " symbolically.
3	For following arguments determine whether the argument is correct or incorrect and explain why?. "Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. There fore, Natasha is a computer science major. "
4	Is the set $\{\{1, 2, 3, \ldots\}\}$ finite? Why?
5	If $\|A\| = 24$, $\|B\| = 15$ and $\|A \cap B\| = 7$, then find $\|A - B\|$
6	Define Mod and Div Functions
7	Draw the function $f(x) = \begin{cases} \|x-1\| & \text{when } x \leq 2 \\ 2x+1 & \text{when } x > 2 \end{cases}$; x is a real number
8	A word over the alphabet 0, 1, 2 is called a ternary word. Find the number of ternary words of length 5 that can be formed.
9	Race cars for a particular race are numbered sequentially from 12 to 115. What is the probability that a car selected at random will have a tens digit of 1?
10	Find the number of ways a four-member committee can be chosen from 10 people

Section B

All Question can be answered. Each Question carries 6 marks (Ceiling : 36 Marks))

11	Using the laws of logic , show that $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology
12	Check the validity of the following arguments. Babies are illogical. Nobody is despised who can manage a crocodile. Illogical persons are despised. --- $\therefore$ Babies cannot manage crocodiles.
13	Using a truth table, prove that $(A \cup B)' = A' \cap B'$
14	A survey conducted recently among 300 adults in Omega City shows 160 like to have their houses painted green, and 140 like them blue. Seventy-five adults like both colors. How many do not like either color?
15	Today is Thursday. What day of the week will it be in 100 days from today?
16	Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 9 & 3 \\ 9 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 4 & 3 \\ 1 & 3 & 1 \\ 7 & 2 & 0 \end{bmatrix}$ Show that $(AB)^T = B^T A^T$
17	Find the number of words with or without meaning canbe made using the letters of the word AGAIN. If these are written as in a dictionary, what will be the 50th word?
18	Two fair six-sided dice are rolled. What is the probability that the sum is either a perfect square or a perfect cube?

Section C

Answer any ONE. Each Question carries 10 marks (1x10=10 Marks))

19	Let A and B be two disjoint finite sets. Then prove that $\|A \cup B\| = \|A\| + \|B\|$.
20	A group consists of 4 girls and 7 boys. In how many ways can a team of 5 members be selected if the team has 1. no girls 2. at least one boy and girl